Samba Gangineni

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Profile

Senior Developer with 5+ years of experience leading high-performance teams and architecting scalable microservices in the Node.js, Java Spring, and Python FastAPI ecosystems. Adept at designing and deploying robust data pipelines using Apache Spark and Scala, as well as developing innovative chatbot solutions leveraging Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) techniques. Proven track record in delivering reliable, scalable applications that drive business growth and operational efficiency.

Skills

Programming: Javascript, Kotlin, Python, Java, React, Flask, Go, SQL, bash, HTML, CSS, HCL
Databases: MongoDB, MSSQL Server
Frameworks & Tools: Temporal, Spring Boot, Spark, TensorFlow, RabbitMQ, Docker, Kubernetes, Helm, Terraform, Pulumi, AWS, Git, Jenkins

Experience

Availity *Feb 2020 – Present — Boston, MA*
Software Engineer IV

- Currently leading an ongoing redesign of a legacy application by integrating Apache Spark pipelines and Scala, significantly enhancing performance and scalability to support a multi-tenant architecture.
- Developed an internal Confluence chatbot for knowledge management—implemented RESTful endpoints using Python FastAPI, integrated Large Language Models with a Retrieval-Augmented Generation (RAG) framework, and leveraged vector databases for efficient search and relevance ranking, significantly reducing time-to-information and boosting internal productivity.
- Designed and implemented a microservices-based, stateless product pipeline with an event-driven architecture on Temporal, reducing dependency on MongoDB.
- Transitioned infrastructure code and customer environments from Pulumi to Terraform, enhancing cloud infrastructure stability.
- Achieved a 5x performance improvement in FHIR data deduplication by redesigning the architecture with Kotlin coroutines to minimize database and RabbitMQ load.
- Streamlined FHIR data ingestion using Temporal and managed CI/CD deployments into EKS with Terraform.
- Spearheaded MongoDB sharding for large-scale healthcare data, regenerating 6 billion FHIR resources for 10 million patients.
- Managed large-scale claims data ingestion using Spark, AWS Lambda, EMR, and Kubernetes for 13 million patients.
- Built a clinical inference model to detect data gaps and created realistic clinical synthetic notes using GPT-2 models.
- Automated terminology updates to enable data normalization and support decision-making during COVID-19.

Diameter Health *Jun 2019 – Aug 2019 — Boston, MA*
Informatics and Engineering Intern

- Developed a transpiler in Golang to convert clinical quality language into JavaScript, automating boilerplate code generation.
- Designed and implemented a synthetic data generator for clinical HL7 messages in JavaScript, accelerating testing and development cycles.

Research

ACM DEBS *2019*

Distributed and Event Based Systems

- Architected a real-time object detection system for LiDAR data using clustering techniques and convolutional neural networks.
- Publication: https://doi.org/10.1145/3328905.3330297

Education

Boston University *MSc in Computer Info Systems* *Grad. Jan 2020*

- GPA: 3.86/4.0
- **Coursework:** Big Data Analytics with Spark, Artificial Intelligence, DataMining & Business Intelligence

R.V.R & J.C C.E *B.Tech in Civil Engineering* *Grad. May 2017*

- GPA: 9.23/10.0
- **Coursework:** Programming with C + Practicum, Database Management System, Computer Programming in CEngg

Awards

2019 ACM DEBS Grand Challenge Winner (Object Recognition)
2018 1st Place in BigRedHacks (ResQu - An SOS Application)
2018 3rd Place in Att Hacks (Imound - An App for the Visually Challenged)